Exposé on Master's Thesis

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Title:	Learning a Dexterous Pinch-and-Thread Skill: Reinforcement Learning of Contact-Rich Nut Threading on a KUKA iiwa14 with a SHARPA Hand
Date:	June 26, 2026

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1 Topic

Threading a nut onto a bolt is one of the most common operations in industrial assembly, yet it remains hard to automate: it is contact-rich, demands sub-millimetre precision, and couples a stable grasp with a controlled in-hand turning motion. Recent learning-based assembly systems built in high-fidelity GPU simulation – Factory [Nar+22] and the force-guided FORGE policy [Nos+25] – show that reinforcement learning (RL) can solve such tasks, nut threading among them. These systems, however, use a parallel-jaw gripper on a 7-DoF Franka arm and start from a part that is already grasped; the policy therefore only ever learns to thread a pre-pinched nut and never has to acquire the grasp itself.At the same time, multi-fingered dexterous hands promise far more general manipulation – in-hand reorientation [CXA21; Han+24], tool use [Ked+26], and bimanual skills such as twisting lids off jars [Lin+24] – but are markedly harder to control because of their high degrees of freedom and rich, hard-to-sense contact [Wei+24]. Pairing a dexterous hand with a contact-rich fastening task therefore remains largely open: no learned policy both forms a multi-finger grasp and performs the subsequent threading on an arm–hand system.This thesis closes part of that gap. The goal is to train a single RL policy that, on a KUKA iiwa14 (7-DoF) arm equipped with a SHARPA dexterous hand, performs a thumb-and-index pinch of an M16 nut and threads it onto a standing M16 bolt. Relative to gripper-based assembly the task adds a reach-and-grasp phase, a perpendicular side-pinch of a low-friction nut, and the requirement to turn the grasped nut down the thread into a very tight success window (centring within 2.5 mm, seating within 0.75 mm, with correct rotation). The relevance is twofold: methodologically it studies grasp-and-manipulate credit assignment and contact sensing on a dexterous hand; practically it is a step toward replacing special-purpose grippers with general dexterous hands for fastening, and in the longer term toward transferring such skills from simulation to reality [Han+24; Tan+24].

1.1 Research questions

The thesis pursues four research questions:

1. Can a single PPO policy learn the coupled grasp-then-thread skill on the KUKA–SHARPA system in simulation, starting without a pre-given grasp?

2. Which contact-sensing and reward-shaping choices are necessary to make the contact-rich grip learnable (true finger–nut contact versus geometric proxies)?
3. How well does the learned skill generalize across nut/bolt sizes (e.g. M10 versus M16), and how far can threading depth and multi-turn rotation be pushed?
4. (Stretch) To what extent does a force-aware end-effector observation improve behaviour in the contact-rich threading phase, and what is needed for a later sim-to-real transfer?

2 Approach

The approach builds on a vendored copy of the IsaacLab *forge* task [Nos+25] – itself built on *factory* [Nar+22] – and carries it from the Franka parallel-jaw gripper to the KUKA–SHARPA system. Figure 1 sketches the setup. The following subsections describe the platform and robot, the task and control, the reward and contact design, the learning method, and the results so far.

2.1 Simulation platform and robot

Training runs in IsaacLab v2.3.2 on Isaac Sim, which simulates contact-rich assembly across many environments in parallel on the GPU. The robot is a KUKA iiwa14 (7-DoF) arm with a left SHARPA hand (22 finger joints). Only the thumb (5) and index (4) are active, driven through a low-dimensional pinch synergy; the middle, ring, and little fingers are frozen curled away to remove irrelevant degrees of freedom and self-collision clutter. The task is realized as a new `forge_kuka` task family with its own gym id, leaving the original Franka forge tasks untouched.

2.2 Task and control

The nut sits physically on the bolt and is laterally constrained by the shaft, so the policy must grip it in order to turn it. The 7-DoF arm is driven by a task-space impedance controller (operational-space control, $\tau = J^\top(\Lambda \dot{w}^* + \mu + p)$), while the fingers are position-controlled. A *static, flange-anchored* end-effector control frame decouples arm control from finger motion, so gripping does not disturb the arm Jacobian. The action space comprises end-effector pose increments (position and orientation) plus the pinch finger degrees of freedom; the yaw action is mapped unidirectionally to the tightening direction.

2.3 Reward design and contact sensing

The reward combines forge's multi-scale keypoint threading reward with a staged grasp shaping: approach → contact → perpendicular side-pinch → turn. Crucially, the threading and engagement terms are gated on a *real, filtered* finger–nut contact signal. Early attempts failed on proxy signals – distance to the nut centre, unfiltered net contact forces – which either mis-matched the geometry or were corrupted by finger–finger self-contact. The breakthrough was to read the filtered pairwise contact force (`force_matrix_w`) of the finger pads against the nut's *rigid-body link*, which separates the true grip from self-contact. Grasp theory (force closure [Ngu86]) and the composition of grasping with in-hand manipulation [Rös+25] provide the conceptual frame for this design.

2.4 Learning algorithm and methodology

Learning uses PPO (rl_games) in an asymmetric actor-critic setup with a privileged critic state and an LSTM, under forge-style domain randomization (control gains, thresholds, action smoothing, friction). Development follows a strict *experiment cycle*: one configuration change per run, one commit per run, results recorded before committing, so that `git checkout` of a run reproduces its documented result. This places the work within the broader line of RL for robotics [KBP13; Tan+24] and, specifically, scaled dexterous arm–hand learning [Pet+23; Lum+24; Sin+25].

2.5 Preliminary results

Across a sequence of experiments the policy reaches roughly 85 % threading success on the M16 nut, once a real, properly filtered finger–nut contact gate and a hand-tuned seated spawn pose replaced the earlier proxy signals. Remaining limits are shallow threading depth, i.e. a single partial turn (the wrist's limited yaw range, with no learned re-grasp), and a grip that registers nearer the knuckle than the pad tip. These points define the open thrusts of the thesis.

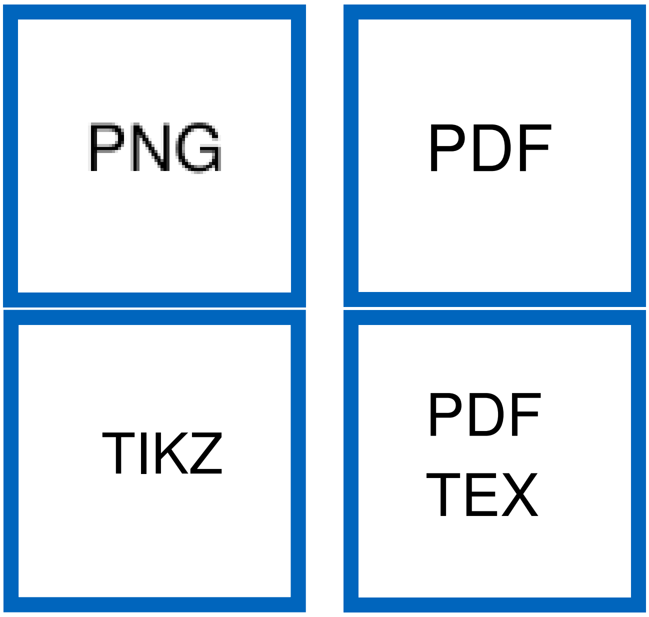

Figure 1: System overview (placeholder figures): the KUKA iiwa14 with SHARPA hand pinches an M16 nut with a thumb–index grip and threads it onto the standing bolt, driven by the staged grasp-then-thread reward.

3 Work Plan and necessary Resources

3.1 Methodology and evaluation

The primary success metric is threading success rate (nut centred, seated to target depth, correctly turned). Secondary metrics are grip quality – contact rate, grip perpendicularity, contact force, and sustained-contact duration – together with episode length and threading depth (nut–bolt distance). Because every configuration change is isolated in exactly one reproducible run, the contribution of each design decision (contact signal, grip gate, spawn pose, end-effector frame, reward weights) can be established by ablation. Comparisons are made (i) against the pre-grasped forge baseline and (ii) across nut/bolt sizes, for which custom M10 assets with SDF colliders were authored. As a stretch goal, a first sim-to-real transfer on the physical KUKA–SHARPA system is planned, time and hardware permitting.

3.2 Resources

The work requires: Isaac Sim with IsaacLab v2.3.2 in a uv-managed environment; a multi-GPU workstation (training runs take hours; remote inspection via WebRTC livestream); the USD assets of the KUKA iiwa14 and the SHARPA hand, and the `forgeUltra` codebase. The optional sim-to-real part additionally requires a physical KUKA iiwa14 with a SHARPA hand and corresponding supervision.

3.3 Schedule

Figure 2 shows the planned sequence: literature and onboarding, baseline reproduction, the grasp-and-thread policy, contact and reward refinement, generalization across sizes, an optional sim-to-real transfer, and final evaluation and writing.

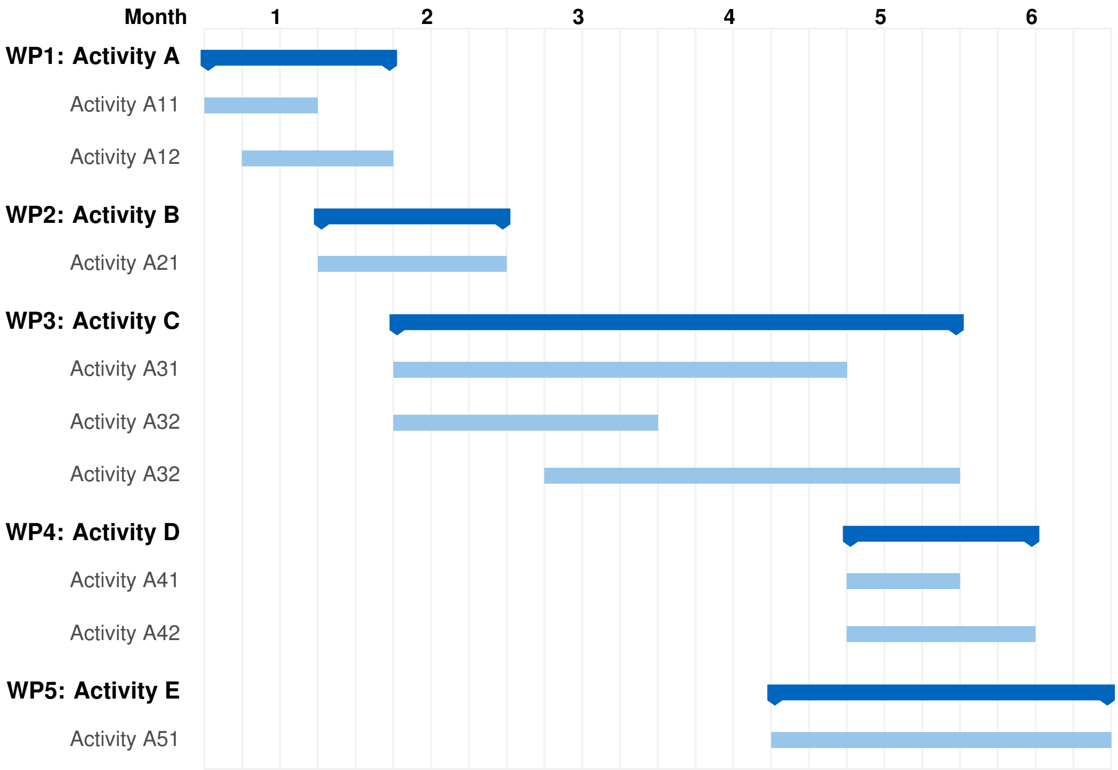

Figure 2: Time schedule.

4 Literature

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